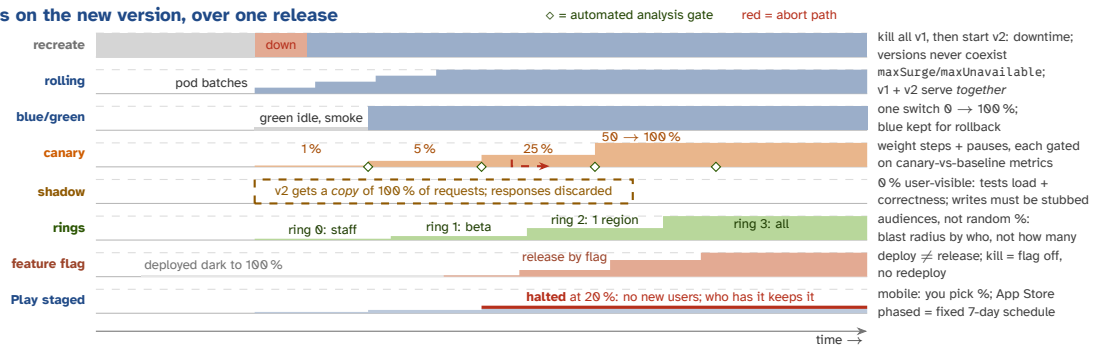


**In one line: Expose a new version to users in steps**, measure each step **against a baseline**, and **promote or abort automatically** — so a bad release hurts 1–5 % of traffic for minutes instead of everyone for hours. **Deploy ≠ release**: the binary can be everywhere while the feature is off.

#### % of users on the new version, over one release



### Rolling update — the default

- K8s RollingUpdate: maxSurge (extra pods, % rounds **up**) and maxUnavailable (% rounds **down**), both default **25 %**, not both 0. 10 replicas:  $\leq 13$  pods,  $\geq 8$  ready. maxUnavailable: 0 = never below capacity.
- Only **readiness** gates progress (+ minReadySeconds); nothing looks at error rates. progressDeadlineSeconds (600) only marks the rollout **failed** — **no automatic rollback**: kubectl rollout undo.
- v1 and v2 serve *at the same time* ⇒ API, schema and messages compatible both ways.
- Recreate** only when versions cannot coexist (singleton, exclusive lock).

### Canary + automated analysis

- Route a **small weight** to v2 (mesh, ingress, Gateway API, ALB weights). No router ⇒ weight  $\approx$  replica ratio (1 of 10 pods = 10 %).
- Compare canary with a **baseline** of the *old* version deployed *at the same time, same size* — not with the big, warm production fleet (caches, JIT, heap differ).
- Golden signals**: latency (p99, successes and errors separately), traffic, errors, saturation; plus restarts/OOM and a business metric (checkout success).
- Kayenta** (Spinnaker): Mann–Whitney U per metric at 98 % confidence → Pass/High/Low; score = % of metrics passing vs *marginal* / pass thresholds; critical metric fails → 0. Needs  $\geq 50$  data points per metric: steps last minutes–hours.
- SLO burn rate**: error budget used per unit time;  $14.4 \times$  over 1h (and 5 min) = 2 % of a 30-day budget ⇒ abort.

### Abort criteria — write them before the rollout

Error rate or p99 worse than baseline beyond a set margin · burn rate over threshold · crash loops / OOMKilled · saturation (CPU, pool, queue depth) · business KPI drop · **no data = fail**, not pass. Abort = weight to 0, canary scaled down; investigate, don't retry.

### Sticky routing

Per-request % sends one user to v1 then v2: flickering UI, lost session, v2-written data read by v1. Route by **user/device id hash**, cookie or header (Flagger A/B mode; ALB target-group stickiness); send *internal* users via a header. Mobile: route by App-Version.

### Example — Argo Rollouts canary

```
apiVersion: argoproj.io/v1alpha1
kind: Rollout
spec:
  replicas: 10
  strategy:
    canary:
      canaryService: api-canary # router splits by weight
      stableService: api-stable
      trafficRouting: { nginx: { stableIngress: api } }
      analysis: # background: whole rollout
        templates: [{ templateName: success-rate }]
      steps:
        - setWeight: 5
        - pause: { duration: 15m }
        - setWeight: 25
        - pause: {} # waits for a human: promote
      # AnalysisTemplate: successCondition: result[0] >= 0.99,
      # interval: 5m, failureLimit: 2 -> abort: weight 0, Degraded
```

### Tools

|               |                                                                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Argo Rollouts | Rollout CRD: canary steps (setWeight, pause, analysis), blueGreen; AnalysisTemplate (Prometheus, Datadog, web, job)                                                  |
| Flagger       | wraps a Deployment; interval, threshold, stepWeight, maxWeight; promote after interval × max/step, roll back after interval × threshold; canary, A/B, B/G, mirroring |
| Spinnaker     | pipelines + <b>Kayenta</b> automated canary analysis                                                                                                                 |

### Shadow, rings, flags

- Shadow / dark launch**: mirror live requests to v2, compare responses offline. Great for rewrites and load; dangerous with side effects (double charge, double email) — stub writes.
- Rings**: staff → beta → one region → all, each baked.
- Flags**: deploy moves *code*, a flag releases *behaviour* — per user, reversible in seconds. Canary the binary, flag the feature.

### Mobile — the store is your rollout controller

|             |                                                        |
|-------------|--------------------------------------------------------|
| App Store   | phased 1/2/5/10/20/50/100 %, 7 days, auto-update users |
| Google Play | staged: any %, <b>halt</b> (no new users), resume      |
| EAS Update  | eas update --rollout-percentage=10 (JS only)           |
| server      | flags / kill switch: the only lever on installed apps  |

### Traps

- ! “K8s rolls back a bad rollout” — it stops; undo is on you.
- ! Canary vs the whole prod fleet instead of a same-age baseline.
- ! 5 % of a quiet hour = too few requests to judge anything.
- ! Shadowing a write path: every payment happens twice.

### Remember

**Small step, same-age baseline, pre-written abort.**